Gamifying of Financial Literacy for University Students

University of Technology, Jamaica

Major Project (PRJ4020)

Student Names:

Alanzo Harris (2210195)

Jordan Bogle (1704233)

Kaydeen Walker (1902433)

Monica Foreshaw (2104888)

Xavier Gardner (2002885)

Supervisor: Dr. Lisa Facey-shaw

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Chapter 1

Abstract

This project, Gamifying of Financial Literacy for University Students, aims to embed financial literacy knowledge in young people through a gaming interface. The project addresses the lack of financial literacy among youth by creating a gaming user interface with various elements to teach them about different aspects of financial literacy for a better future. The key processes include requirements gathering, system design analysis, implementation and testing. The expected deliverables are a working prototype, a technical report and a user manual. This will have academic, practical and social benefits.

Background

In the digital age, financial literacy has emerged as a crucial life skill for economic stability and proper decision-making. However, numerous studies reveal that young people continue to demonstrate significant gaps in financial knowledge and competency, often due to limited formal education on the topic and low motivation to engage with traditional teaching methods (Estima & Peguinho, 2024). The deficiency in financial understanding not only affects individual economic well-being but also has broader societal implications, influencing saving behavior, debt management and long-term financial security. As the financial landscape becomes increasingly complex, the need for innovative and accessible educational strategies to improve financial literacy among youth has become a global priority (United Nations Development Programme, 2023).

Gamification has gained increasing attention as a pedagogical approach capable of transforming learning experiences. By integrating elements such as points, levels, challenges and rewards, gamification fosters engagement and motivation through playful competition and goal achievement (Diaz & Estoque-López, 2024). Empirical research has demonstrated that gamified learning environments can significantly enhance student motivation, knowledge retention and cognitive engagement compared to traditional instruction (Ruiz-Carhuamaca et al., 2024). In the context of financial education, game-based learning has proven particularly effective, as it allows learners to simulate real-world financial decisions in risk-free environments, promoting practical understanding and behavioral change (Estima & Peguinho, 2024).

The increasing prevalence of digital learning platforms and mobile applications presents new opportunities to embed financial literacy education within the technologies that young people already use. Studies indicate that interactive and gamified applications can not

only improve conceptual understanding but also sustain long-term behavioral shifts toward responsible financial practices (Cannistrà et al., 2024). Ruiz-Carhuamaca et al. (2024) emphasize that mobile gamified learning tools can bridge the gap between knowledge acquisition and practical application by allowing learners to engage in scenario-based challenges that mirror real financial contexts. Similarly, Lisana et al., (2025) highlights that Generation Z's preference for interactive, visually engaging and technology-driven learning methods makes gamified financial education particularly effective for this demographic.

Despite these advancements, existing gamified financial education tools often lack adaptability, interactivity, and localized content relevant to young users' everyday experiences (Szecsenyi & Monauni, 2024). Many applications focus primarily on abstract financial concepts without providing sufficient feedback or real-world simulation to reinforce decision-making skills (Bitrián et al., 2021). The United Nations Development Programme underscores the potential of gamified education to build financial resilience through play but notes the need for contextually designed solutions that combine educational rigor with user-centered engagement strategies (United Nations Development Programme, 2023).

This gap highlights the necessity of developing gamified learning solutions specifically designed for youth that not only convey financial knowledge but also cultivate the motivation and confidence to apply it in real-life situations. By merging financial education with engaging gameplay and interactive learning design, this project seeks to contribute to the growing body of research that explores how gamification can promote financial literacy, empowerment and long-term economic resilience among young people.

Problem Statement

Financial literacy is a critical life skill, yet studies consistently show that young people often lack the knowledge and decision-making ability to manage money effectively (Lusardi, 2019; Mao et al., 2023; Trivedi & Gupta, 2024). A lack of financial knowledge can lead students to make decisions that can negatively affect their future. Bad investment and budgeting can lead students to incur debt that can lead to bankruptcy. Research highlights that gamification and game-based learning approaches can improve motivation, knowledge retention, and behavioural change in financial education (Estima & Peguinho, 2024; Ruiz-Carhuamaca et al., 2024). However, there is limited availability of digital learning platforms that combine interactivity, practical simulations, and feedback specifically tailored for youth that will make learning about money more engaging, practical and effective.

Research Question

1. To what extent are University of Technology, Jamaica students financially literate?
2. How effective is gamification in improving financial literacy among students from the University of Technology, Jamaica?

Purpose of Study

This study aims to assess how gamification of financial literacy can be used to motivate and educate students. By ascertaining the current level of financial literacy of university students in Jamaica, we aim to investigate how effective the use of game based elements like rewards, challenges and levels, can improve their knowledge of finance.

Significance of the Study

This research is significant as it dives into how gamification can assist university students that are not financially literate, which can lead to overspending, failure to save, bankruptcy, foreclosure of housing, and unsustainable debt burden. The findings that will be gathered from this study are expected to contribute in building financial literacy in university students using gamification, which will make learning more fun and interactive. The research will contribute to the body of the literature by investigating gamification as a teaching method for enhancing university students' financial literacy.

Limitations

This study will rely on self reported data which may introduce bias. The sample will be a relatively small number of students of UTECH, Jamaica, hence it may limit the generalizability of the study to represent all universities in Jamaica or globally. Additionally, Lack of a proper budget will prevent the creation of a fully developed application, therefore, a workable prototype will be created for the purposes of the study.

Delimitations

This study will only include university students currently enrolled at the University of Technology, Jamaica. Additionally, the study will only focus on fundamental personal finance topics such as basic investing, budgeting, debt management, and saving. Due to time restrictions, the study cannot focus on all concepts; however, further investigations will be done, gaining expert opinion on which concepts are pertinent. The scope of this project is to measure learning outcomes and user engagement/satisfaction with the prototype during a short testing period. It will not attempt to track long-term behavioral changes or the participants' actual spending or saving habits. The focus will be on the learning outcomes of the gamified approach.

Definition of Terms

Game-based elements: Game based elements include point systems, levels, badges, leaderboards, challenges, onboarding, and engagement loops. These components serve to track player progress, measure skill, facilitate interaction, and maintain engagement within gamification systems (Science Direct, n.d.).

***Pedagogical approach*:** Pedagogy in education is the study of teaching methods that educators use to help students meet learning objectives. Pedagogical approaches vary depending on the subject matter, students' education level and classroom dynamics, but three elements are always present: the educator, the learner and the subject matter (Augusta University, 2024).

Gamification: Gamification is the art of applying game-like elements and principles to non-gaming contexts (Anuradhani et al., 2023).

***Financial literacy*:** financial literacy can be defined as the set of skills needed to handle money wisely, invest effectively, and plan for a comfortable future (Fernando, 2025).

Generation Z: The generation of people born between the late 1990s and early 2010s and following Generation Y, noted in particular as the first generation to grow up in the era of widespread use of digital technology (Oxford English Dictionary, 2023).

Proposed Methods

The methodology used for this study will be a mixed-methods approach, integrating both qualitative and quantitative techniques to provide a strong understanding of how a gamification approach can improve financial literacy among university students. We will gather quantitative data on the financial literacy level of university students through the dissemination of a structured questionnaire. This will measure their existing level of financial

literacy having not experienced the gamification approach which will give a baseline level in this target group. Qualitative data will also be obtained using interviews focused on the respondents view of their personal financial status and their opinions on the financial decisions that they have made since becoming adults. Lecturers at the University of Technology, Jamaica will also be interviewed to ascertain which financial concepts are most pertinent for Jamaican University students to learn. Additional qualitative data will be gained from user feedback on an early prototype of the gamified learning tool that will be developed. This will bring insights on the motivation, engagement and ease of learning achieved by the gamified approach.

This study will use purposive sampling to gather data. Content experts will be interviewed to gain insights on the field. Additionally, stratified convenience sampling will also be used to gather information. Participants will be gathered from across the different faculties of the University of Technology, Jamaica. However, the number of students from each faculty will be in proportion to the percentage of the total student body that each faculty represents. This will allow us to gain a full picture of the overall financial literacy levels of the students body.

The Quantitative data will be analysed with descriptive and inferential statistics to summarize and draw conclusions from the data. This will be combined with a thematic analysis of the qualitative data, where trends and patterns will be identified to gain a deeper understanding of the target audience. This approach to research will provide a comprehensive grasp of the level of financial literacy among university students and how gamification may improve their knowledge in this area.

Chapter 2 – Literature Review

Introduction

Financial literacy is an essential life skill for all young people and there is a growing concern that young people may be falling behind in this area. According to Klapper et al. (2016), only 33% of Jamaican adults are financially literate. According to Lusardi (2019), findings from the FLAT World project, shows the financial literacy level across 15 countries worldwide is 30%. This is described as a "crisis level". The title FLAT World actually means Financial Literacy Around the World (Lusardi & Mitchell, 2011). It is a project that combines data from fifteen participating countries in an attempt to ascertain the level of financial literacy around the globe. Urban & Valdes (2022) mention that between the period of 2009 and 2021 the United States of America has seen a 26% decline in financial literacy rates from the age group 18-29. There seems to be a need for new solutions to address this issue and this study aims to ascertain if gamification can be a useful tool in tackling this issue.

There are different variables that affect financial literacy. Gupta & Trivedi (2024) found that 56% of Indian university students were financially literate. However, they posit that education level is a major contributing factor in the respondents' basic financial knowledge. Mao et al. (2023) found that around 36% of their respondents were financially literate after doing an objective assessment. Their assessment found that parents have the most perceived influence on college students' financial literacy. According to Lusardi & Mitchell (2014), there are various other empirical patterns in financial literacy that are consistent across countries. Financial literacy differs greatly based on socioeconomic status, including employment and income type. Self employed and working individuals outperform those who are unemployed, while lower-income persons generally have lower levels of financial literacy (Lusardi & Mitchell, 2014). Additionally, Lusardi (2019) stated that age is

an essential factor in determining financial literacy, being lowest among the old and the young. In the United States, less than 30% can accurately answer basic financial literacy questions by the age of 40, despite having made significant financial decisions far earlier (Lusardi, 2019). Furthermore, in Switzerland, only 45% of individuals aged 35 and under can accurately respond to basic financial literacy questions (Lusardi, 2019). Elderly individuals who have previously made financial decisions often struggle to correctly answer these same questions, indicating a limited capacity for learning from financial experiences(Lusardi, 2019).

According to Chiu (2025), a gamified approach to financial and investment education resulted in improved the financial knowledge of their subjects by 31%. Gamification also improved engagement by 87% and reduced learning anxiety by 78%. Nasrun et al. (2024) was able to demonstrate a direct correlation between gamification and an increase in the enjoyment and engagement of students while learning about financial topics.

This review will discuss three main themes: Financial Literacy and Its Importance, Gamification as a Learning Tool and Gamification in Financial Education. The research gaps in this area of study will also be addressed.

Understanding Financial Literacy

According to Fernando (2025) financial literacy can be defined as the set of skills needed to handle money wisely, invest effectively, and plan for a comfortable future. Some of the key components of financial literacy are budgeting, debt management and investing. Budgeting is one of the most foundational aspects of financial literacy and is something that is useful for university students as they learn to be independent adults. Budgeting provides benefits such as fixing spending habits, leaving resources available in an emergency and it can open up financing opportunities (Cote, 2022; Knisley, 2023). Knowledge of budgeting

could greatly benefit young people in Jamaica. According to Market Research Services Limited (2023), students demonstrated strong conceptual recognition of budgeting but weaker practical application, particularly in distinguishing fixed versus variable expenses. Debt management can be defined as "the process of planning your debt liabilities and repayments" (Smith, 2024). As many students will have to take student loans to complete their studies so knowledge of debt management would be vital for university students. Finally, investing is also a vital area for university students to be knowledgeable in. Investing is also the area in which students know the least. According to Mao et al. (2023) only 35.89% of university students correctly answered the objective financial knowledge question regarding investing and only 34% of students reported learning about investing from home. In investing, the sooner one learns about investing, the better off they will be due to the fact that most investments are left to mature over years.

The advantages of being financially literate are enormous. Lusardi and Mitchell, (2014) discusses the empirical evidence pointing to how much better off financially literate people are compared to those who are lacking in financial knowledge. They go on to link financial literacy with different economic outcomes. The study mentions that the people who are least financially literate are youths and elderly people and this makes them the most prone to making financial mistakes. These can include bad investments, frivolous purchases or falling victim to financial scams. Being financially savvy can protect people from making these types of mistakes. It also goes to say that the more financially literate someone is, the more likely they are to undertake retirement planning. This often entails making smart investments that will benefit someone once they reach retirement age. This link has been replicated in many studies and validates how important financial literacy is.

Market Research Services Limited (2023) finds that the factors which influence financial behaviour the most are age, banking status and financial knowledge. It posits that greater financial knowledge can lead to healthier spending habits and a higher likelihood of saving. OCED (2024) analysed the results of the Programme for International Student Assessment (PISA) taken by hundreds of thousands of students across 20 countries and empirically showed the link between financial literacy and better financial behaviours and outcomes. The proficiency of students was ranked from levels one to five. It was found the students at proficiency levels four or five were 72% more likely to have saved to an account or at home and were 50% more likely to compare prices of products compared to those who were at proficiency level one. It is clear that being financially literate gives young people the tools to manage their debts, avoid financial mistakes and plan for their futures.

Global and Local Findings on Financial Literacy Levels Among Youth/University Students.

It is crucial for young people to comprehend finance in order to effectively select and manage financial products.. However, Rodríguez-Correa et al. (2025) found that previous studies indicate the financial literacy level among youth is moderate in Malaysia, while it is low in most other parts of the world. To be particular, the study conducted at Universiti Kebangsaan Malaysia (UKM) found that undergraduate students had a moderate level of financial literacy (Mohd Noor et al., 2024). The majority of students struggle to save their present student loans primarily caused by excessive spending on academics or personal lifestyle (Mohd Noor et al., 2024). Additionally, a study was conducted on accounting students which discovered that financial education in accounting students and behaviour were moderate (Rodríguez-Correa et al. ,2025). Furthermore, Erden (2024) conducted a study and computed the participants' financial literacy scores (92.2% of participants aged 15 to 26),

with 74% of university students scoring between 10 and 17 points, indicating a medium level of financial literacy (Erden, 2024). The most common score was 15 points (15.5%), followed by 13 (13.8%) and 12 (13.3%). A small percentage of participants scored very low (1-5 points, 3.3%), while just 3.3% received the highest score of 19 points. Similarly, Ergün (2017) conducted a study on the level of financial literacy among university students across eight European countries and reported an average score of 14.4 out of 20, suggesting a medium level of financial literacy, with a mean proportion of accurate responses of 72.2%.

Locally, Market Research Limited (2023) reported that Jamaican youth usually grasp budgeting, insurance and basic loan concepts; however, smaller out of school youth and unbanked youth have less knowledge. Nearly 44% of in school and 49% of out of school young people believe that credit cards are not the same as bank loans. Less than half comprehend compound interest (47% in school and 43% out of school), emphasizing the importance of targeted financial education. Using the elements; knowledge, practice, education and attitude, school youths record an index of 36.86 (67.02%) points out of 55 points in financial literacy, while out of school youths record a score of 37.08 (67.42%) (Market Research Services Limited, 2023). Based on the findings, it can be concluded that Jamaican youth have a moderate level of financial literacy. In addition, financial knowledge is statistically significant ($p < 0.05$) in predicting financial practice, suggesting that higher levels of knowledge correlate with better financial behaviors among students. However, in general, youths display poor financial habits (Market Research Services Limited, 2023). According to the research, globally and locally, young people have a medium level of financial literacy, they may have some knowledge but lack the capacity to put it into practice. This demonstrates that there is room for improvement in financial literacy among young people.

Financial Literacy, Decision-Making and Financial Stability

Individuals encounter difficult financial decisions with potentially long-term effects at every phase of their lives. As young individuals enter into adulthood, they make decisions concerning tuition, houses, vehicles, as well as managing credit card, insurance and budgeting (Lusardi & Mitchell, 2023). The transition implies that people must now make good decisions about investing, saving etc. Financial literacy among young people is critical to ensure that they make sound financial decisions. According to studies, students with less financial awareness are more likely to engage in behaviours that have academic consequences (Rodríguez-Correa et al., 2025). Furthermore, people who lack financial literacy are more likely to make poor financial decisions, which harm their financial stability (Mohd Noor et al., 2024). For instance, research stated that students compute compound interest and, as a result, make errors when evaluating financial information(Rodríguez-Correa et al., 2025). Also, insufficient money management is frequent, and few students grasp limits on credit, interest rates and loan repayment liabilities. Students also struggle with budgeting, but they are the least interested in it globally (Rodríguez-Correa et al., 2025). This fosters unhealthy spending behaviour, coupled with low financial literacy, leading to improper management of finance and unsustainable lifestyle. The mere fact that so many individuals lack financial understanding not only reduces their capacity to maximize their resources, but also leads to macroeconomic concerns such as financial crises like subprime mortgage debacle, economic instability like debt etc. (Lusardi & Mitchell, 2023).

Additionally, Rodríguez-Correa et al., (2025) suggests that early exposure to financial literacy education results in responsible financial choices later in college. This confirms that gaining financial information early in life significantly influences people's financial decisions and outcomes. According to Erden (2024),the higher the financial literacy, the better one's decisions are at saving, investing and managing long-term financial goals. This provides the

foundation for personal financial security (Erden, 2024). Based on the findings, it can be concluded that low financial literacy can impair decision making, promote taking risks with finances, and negatively impact an individual's long term financial security and welfare; however, developing financial literacy can help make informed decisions.

Limitations of Traditional Teaching Methods

It is essential to educate people about personal finances. However, previous research indicates that traditional methods to teach finance may lack experiential learning aspects that assist students in integrating practice and theory (Suleiman, 2024). Traditional teaching techniques frequently fail to include students actively in the process of learning (Falasi, 2024). Weak information reception might result in lack of engagement and critical thinking skills. Furthermore, this strategy may not always accommodate different methods of learning, which may leave certain students behind (Falasi, 2024). As a result, numerous individuals must learn financial literacy from their family. According to research, 35 of 65 respondents believe that the influence of family is the most influential component in building financial literacy. Family impact accounts for 38% of factors that affect financial literacy (Mohd Noor et al., 2024). Additionally, Mohd Noor et al. (2024) found that Universiti Sultan Zainal Abidin (UniSZA) students who constitute a small subset of Malaysian youth, achieve a moderate level of financial literacy, with family being the most influential factor. However, family influence can be problematic. Another study found that family members lacking financial literacy might make harmful decisions which impact not just themselves, but their families, and society at large (Ergün, 2017.).

There are also substantial differences in education level. Although sixty five percent of individuals with college degrees or more scored a perfect score on the Big Three, which is three primary questions asked regarding financial literacy, just eighteen percent of secondary

school dropouts achieved this level (Lusardi & Mitchell, 2023). Nonetheless, more than one third of respondents, including those with a college degree, were unfamiliar with one or more of the three questions. This shows that higher education alone is inadequate to teach people financial literacy (Lusardi & Mitchell, 2023). This emphasizes the notion that acquiring financial knowledge involves further investment that is not currently included in traditional education. Lusardi & Mitchell (2023) discovered significant findings from earlier studies that financial literacy is poor even in countries with high incomes (such as Italy and Japan) and those with a good educational system like Sweden. Furthermore, Mohd Noor et al. (2024) suggest that academicians, governments, agencies, individuals, and non-governmental organizations (NGOs) should raise awareness about financial literacy by developing more efficient financial awareness programs and encouraging students to practice good money management in order to improve their financial literacy level.

In addition, financial literacy for young people is increasingly acknowledged as crucial, but its implementation is difficult in emerging economies due to resource constraints, public regulations, and educational approaches (Rodríguez-Correa et al., 2025).To address these challenges, modern teaching approaches emphasize student centered learning(Falasi, 2024). Project based learning, collaborative learning and the utilization of technology are common approaches. These strategies enable students to actively participate in their education, encouraging problem solving skills, critical thinking and inventiveness (Falasi, 2024). Technology integration, such as whiteboards with interactive features, software for education, and online tools, enhances the engaging and easily accessible nature of learning (Falasi, 2024).

Educational Advantages of Gamification

Gamification has been recommended by several authors to improve the financial literacy of university students (Nasrun et al., 2024; UN Sustainable Development Group, 2023; Zainuddin et al., 2020). Research consistently demonstrates that gamification improves learning by boosting motivation, engagement, and knowledge retention.

Users' motivation is enhanced through elements such as a points system, badges, leader boards for tracking peers to incite competition, and progress monitoring like streaks, which provide learners with instant feedback and a sense of accomplishment. These mechanisms appeal to both intrinsic and extrinsic motivational factors, prompting learners to remain devoted to their tasks for extended durations.

Furthermore, engagement also increases as game-like features render learning experiences more interactive, immersive, and enjoyable, alleviating feelings of boredom or cognitive strain. Studies have shown that when students are deeply engaged, they are more prone to participate, collaborate, and persevere through difficult material. Smirani and Yamani (2024) found that these elements increased learner engagement by an average of 25%, demonstrating the strong influence of game mechanics on student participation. Platforms like Kahoot!, Classcraft, and Badgeville further support engagement by offering interactive quizzes, collaborative missions, and progress-tracking systems that encourage sustained involvement.

Finally, gamification fosters better knowledge retention by integrating repetition, rewards that are linked to completing tasks, causing the user's brain to associate the problem with the pleasure of getting rewarded, and active problem-solving, all of which support long-term memory development. Consequently, gamified learning environments advance deeper comprehension and sustained academic success across various educational settings.

Gamification in Higher Education

Gamification has become an increasingly influential instructional strategy in higher education, where universities aim to improve student motivation, engagement, and learning outcomes through game-based elements such as points, badges, progress bars, and leader boards.

Research shows that when these mechanics are thoughtfully integrated into academic contexts, they help increase student participation, persistence, and intrinsic motivation by making learning activities feel more interactive and rewarding (Deterding et al., 2011; Zainuddin et al., 2020).

Within university courses, gamification is applied through online learning management systems, classroom exercises, quizzes, and collaborative challenges that encourage students to monitor their progress, compete in healthy ways, and engage more deeply with course content (Kapp, 2012). Studies further reveal that gamified systems in higher education promote self-regulation, enhance knowledge retention, and support mastery of complex concepts, particularly in digital and blended learning environments (Ibáñez & Delgado-Kloos, 2018).

However, evidence also cautions that poorly designed gamification can undermine learning by creating unnecessary competition, reducing intrinsic motivation, or distracting from core objectives (Hanus & Fox, 2015). Therefore, successful gamification in higher education requires balancing meaningful rewards with pedagogical alignment to ensure that game elements support, not replace, the learning process.

Gamification in Financial Education

A number of tools have been developed that use gamification to address various aspects of financial literacy. Some platforms focus on building knowledge and skills through structured lessons. For example, Zogo teaches financial concepts using quick quizzes and rewards users with real gift cards for correct answers, making learning feel rewarding. Similarly, Wall Street Survivor allows users to learn about investing in a safe way by offering a simulated stock market where they trade with virtual money while earning badges for completing missions. Other tools use game mechanics to tackle real-world financial habits. Fortune City turns the sometimes-tedious task of budgeting and expense logging into a fun city-building game, where tracking transactions actually grows the user's virtual metropolis. Likewise, Qapital makes saving money feel like a challenge by setting up automated "if this, then that" rules that help users save small amounts regularly. Finally, apps like Habitica take a broad approach, treating all real-life goals, including financial tasks, as role-playing game (RPG) quests, allowing users to earn experience and level up their avatar as they achieve their money goals. These examples demonstrate how game design is successfully used to make complex, often stressful financial tasks more engaging, motivational, and accessible.

Gamifying financial education has been proven to be highly successful in making learning more engaging and easier. Adding game elements to financial education can lead to real improvements in understanding. One study found that in a gamified workshop, there was a 31% improvement in financial knowledge, an 87% boost in engagement, and a 78% reduction in learning anxiety (Chiu, 2025). Along with learning, the game aspects also made the class much more interesting, which increased student participation. This participation led to a behavioral change, as 72% of participants applied their knowledge to simulated investments, while another 46% felt motivated enough to enter the investment market for the first time. The success of financial gamification is not limited to just knowing more about

financial topics; it can also influence real-world personal finances. By using fun, competitive elements like digital awards, mobile finance games have been shown to encourage users to save an average of 25% more than those who don't use the games (Bayuk & Altobello, 2019). This suggests that game elements in financial education can lead to higher user motivation to form better real-world financial habits.

Research Gaps

Despite extensive global research on financial literacy and the growing body of work on gamification in education, several critical gaps remain, particularly for the Caribbean context.

Lack of Localized (Caribbean/Jamaican) Research

Most existing studies on financial literacy and gamification focus on countries such as the United States and Malaysia. While these findings are valuable, they do not reflect the unique cultural, economic and educational realities of Jamaican youth.

Local studies (e.g., Market Research Services Limited, 2023) show that Jamaican young people have only a moderate level of financial literacy, with specific weaknesses in applying financial knowledge, yet no research explores whether gamification could improve this.

Few Studies Focus Specifically on University Students

Much of the literature either evaluates general youth populations, secondary school students or adults. However, university students face unique financial challenges: taking loans, budgeting independently, managing credit and preparing for early investments.

The reviewed research shows that young adults worldwide often miscalculate interest, misunderstand credit and struggle with budgeting, yet very few studies test gamified financial learning specifically among university students.

Limited Intersection Between Gamification and Financial Literacy in Developing Regions

Although global findings strongly support gamification's ability to improve engagement, motivation and knowledge retention, there is a lack of evidence on:

- How gamification performs in resource-limited educational settings.
- Whether gamified financial education can address cultural barriers such as family influence, low exposure to investment knowledge or limited access to financial products.

Absence of Jamaican-Based Gamified Prototypes

Numerous financial apps and platforms like Zogo, Habitica, Fortune City and Wall Street Survivor have demonstrated success internationally.

However, none of these tools were designed for Jamaican financial realities, such as local loan structures, NHT/NIS concepts or common cultural misconceptions about credit and saving.

How This Study Fills the Gaps

This research directly addresses these gaps by:

- Developing and testing a gamified financial literacy prototype designed specifically for Jamaican university students.

- Measuring the engagement, usability and effectiveness of gamification in a Jamaican context.
- Providing localized insights into whether gamification can strengthen financial knowledge and motivate better financial behaviour among young adults.

Summary

This literature review explored three major themes: Financial Literacy and Its Importance, Gamification as a Learning Tool and Gamification in Financial Education, to understand the potential value of a gamified solution for Jamaican university students.

Across global and local studies, financial literacy was found to be moderate at best, with significant gaps in practical financial skills such as budgeting, debt management and investing. Young people often recognize concepts but struggle to apply them in real-life situations. Low financial literacy contributes to poor financial decisions, academic challenges and long-term financial instability.

The review also showed that traditional teaching methods are often ineffective because they lack engagement, interactivity and experiential learning. Modern students respond better to interactive and technologically integrated approaches.

Gamification emerged as a strong alternative instructional method. Research consistently shows that game elements: points, badges, challenges and rewards significantly increase motivation, engagement, enjoyment and knowledge retention. In higher education, gamification encourages persistence, self-regulation and deeper learning when designed well. In financial education specifically, gamification has led to measurable improvements, such as increased savings behaviour, greater investment confidence and reduced learning anxiety.

Overall, the literature highlights the need for more engaging, practical and student-cantered ways to teach financial literacy, particularly in developing regions. This creates a clear opportunity for a localized gamified learning tool tailored to Jamaican university students, who face financial pressures but lack accessible, interactive financial education resources.

This study responds to that need by developing a Jamaican-based prototype and evaluating its potential to strengthen financial learning outcomes, making financial education both effective and engaging.

Chapter 3 - Methodology

Research Design

A mixed methods approach will be utilized in this study, combining both quantitative and qualitative research designs to investigate financial literacy among students at the University of Technology, Jamaica. The survey will be disseminated in two parts with the same questions will be used; a pre-survey will be created and distributed to students to determine their financial level; and a post-survey will be distributed after students have interacted with the prototype to determine whether their financial literacy level has improved and to gather feedback on the game's usability. In addition, an interview will be conducted with content experts from the discipline to evaluate relevance and accuracy, as well as to gain deeper insights into the financial literacy level of students.

Sampling

We will engage in the following sampling methodologies:

- Stratified Convenience Sampling – We will first divide the population into subgroups of the faculties then use convenience sampling to select participants from each stratum to get a better representation across all faculties.
- Purposive sampling – We will be interviewing content experts of the discipline in order to gain their expert knowledge.

Data Collection Instruments

Questionnaire

The questionnaire will follow a pre-test post-test design to assess the changes in their knowledge of financial concepts, their attitudes and behaviours toward their finances. The same questionnaire will be administered before and immediately after participants complete the financial literacy course prototype. This will allow us to ascertain if any changes in their financial literacy level came from their interaction with the prototype.

The questionnaire will be divided into three (3) categories:

- Financial Knowledge: Measures participants' actual understanding of financial literacy concepts.
- Financial Attitude and Behaviour: Measures self-reported confidence and the frequency of positive financial habits. It will use a five (5) point scale (1-Strongly disagree to 5-Strongly agree). This will assess the impact on behavioural intention and attitude.
- Gamification Effectiveness: This will measure the participant's view of the tool's motivational components, such as rewards, competition, and overall engagement.

Interview

The interviews will provide an external expert perspective to validate the content accuracy, pedagogical effectiveness, and real-world applicability of the gamified tool. This helps confirm that the observed positive outcomes (as measured by the questionnaire) are based on sound financial principles.

The interview guide will be structured around three areas:

- Content Validity: Expert opinion on the accuracy of the financial concepts presented will be obtained, as well as whether the game adequately addresses key risk areas for students.
- Evaluation: Experts will be asked to critique the effectiveness of the gamified methodology as a teaching tool compared to traditional instruction, addressing its potential for long-term habit formation.
- Scalability: Questions will address the tool's strategic value, including its potential for wide-scale integration into educational institutions and its role in addressing the financial literacy gap.

Procedures

The procedures outline the step-by-step process that will be followed to collect, analyse and validate data for this study. These procedures ensure consistency, reliability and ethical compliance throughout the research process.

Preparation Phase

Before data collection begins, the following steps will be completed:

- Develop and pilot-test the questionnaire to ensure clarity and reliability.
- Prepare the expert interview guide and confirm participating experts from the financial literacy or education discipline.
- Obtain ethical approval from the University of Technology, Jamaica's Ethics Committee.
- Secure informed consent forms for both student participants and expert interviewees.

Recruitment of Participants

- A stratified convenience sampling approach will be used to select university students from the University of Technology, Jamaica.
- Participants will be invited through class groups, campus announcements and email communication as well an in person invitations.
- Students will receive a description of the study, including purpose, expectations, confidentiality, and voluntary participation.
- Experts for interviews will be recruited based on their qualifications in finance.

Administration of the Pre-Survey

- Participants will first complete the pre-survey questionnaire, which measures financial knowledge, financial attitudes and behaviours and baseline perceptions.
- The survey will be administered online through a secure platform (e.g., Google Forms).
- Students will be instructed to complete the questionnaire independently and honestly.

Interaction with the Gamified Prototype

- After completing the pre-survey, participants will gain access to the financial literacy gamified prototype.
- They will be given a fixed duration (e.g., 20–30 minutes) to explore and complete the game activities.
- The prototype experience will be self-paced to simulate real-world usage and minimize researcher influence.

Administration of the Post-Survey

- Immediately after using the prototype, students will complete the post-survey, which contains the same financial knowledge and attitude questions as the pre-survey.
- Additional questions evaluating gamification effectiveness, such as motivation, engagement and usability will also be included.
- This enables direct comparison between pre and post-interaction results to measure learning gains and behavioural impact.

Conducting Expert Interviews

- Semi-structured interviews will be held with content experts.
- Interviews will be conducted in person or via online meeting platforms, depending on expert availability.
- Each session will be audio-recorded (with permission) and transcribed for analysis.
- Experts will review the prototype and provide insights on:
    - content validity
    - pedagogical effectiveness
    - scalability and long-term usefulness

Data Management and Analysis

- Quantitative data will be analysed to compare pre and post-test scores and determine changes in financial knowledge and attitudes.
- Interview transcripts will undergo thematic analysis to extract expert insights and validate findings.
- All data will be stored securely using encrypted digital folders accessible only to the research team.

Integration of Findings

- Quantitative survey results and qualitative expert feedback will be synthesized.
- Agreement between findings from the questionnaires and expert interviews will be used to identify the strengths of the gamified tool. Any differences between the findings will be used to identify weaknesses and guide improvements to the design.
- This combined analysis supports a more comprehensive understanding of the prototype's effectiveness.

Ethical Considerations

This study will adhere to established ethical principles to ensure the protection, dignity, and rights of all participants involved in researching the effects of gamification on financial literacy.

Informed Consent

Participants will be provided with clear information informing them of the purpose of study which is to examine the outcome of gamification of financial literacy of university students. Additionally, the consent form shall provide information on the nature of their participation in the study, which will include the completion of a questionnaire and interaction with a prototype of the learning platform. Participants will also be informed of the duration of their participation in study and any apparent risk or benefits before consent is provided.

Right to Withdraw

Participants will be informed that they may withdraw from the study at any time without being negatively impacted. Choosing not to participate or withdrawing will not affect participants academically, financially, or professionally.

Protection from Psychological and Financial Harm

The study will not collect sensitive personal financial data such as income levels, bank account details, or debt amounts. Instead, the questions will focus on financial knowledge, behaviours, and attitudes rather than personal financial status. The gamified activities will be designed to be educational and transparent, ensuring that participants do not feel pressured to make real financial decisions based on the study.

Confidentiality and Anonymity

All personal and financial-related data will be treated with strict confidentiality. No information that can be used to identify the participants will be included in published findings. Participants will be assigned unique identifiers to anonymize responses and data will be stored secure files accessible only to the research and supervisor.

Ethical Approval

The research will be reviewed and approved by the relevant academic supervisor or institutional ethics committee prior to data collection.

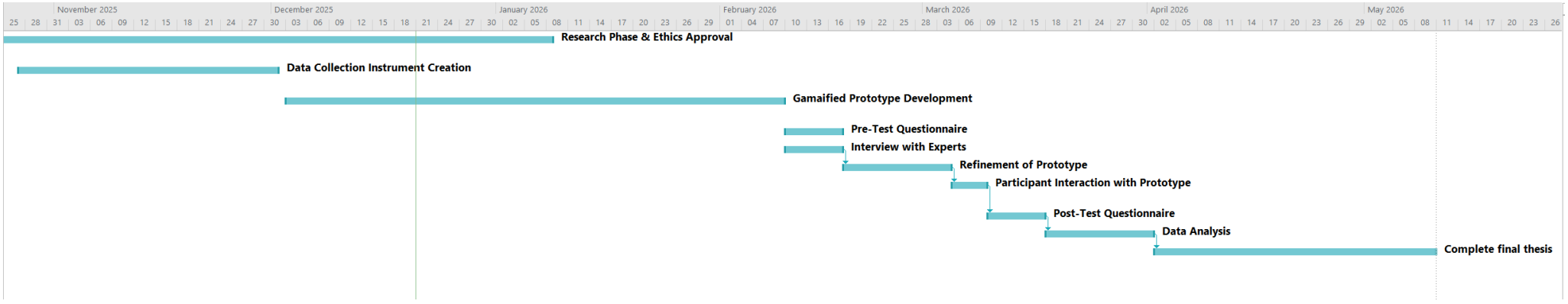
Methodological Limitations

The methodological limitations are as follows:

- Due to the short turnaround between the pre-test and post-test questionnaire's, the study does not assess the long-term knowledge retention capabilities of a gamified platform
- Due to time constraints, the study is limited to the basic concepts of financial literacy (budgeting, saving, debt management, investing). This limits the scope of the project and more advanced financial concepts had to be omitted.
- Some of the data that is being collected are self-reported data from participants. This data may be influenced by the various biases held by participants.
- Our choice of a stratified convenience sampling method was made to increase the diversity and improve representation among our participants. However, with the focus being on students of the University of Technology, Jamaica, and the relatively small sample size, this may limit the generalizability of the study to represent all universities in Jamaica or globally.

Project Timeline

Task Name	Duration	Start	Finish
Research Phase & Ethics Approval	70 days	Fri 03-10-25	Thu 08-01-26
Data Collection Instrument Creation	26 days	Mon 27-10-25	Mon 01-12-25
Gamified Prototype Development	49 days	Wed 03-12-25	Mon 09-02-26
Pre-Test Questionnaire	6 days	Tue 10-02-26	Tue 17-02-26
Interview with Experts	6 days	Tue 10-02-26	Tue 17-02-26
Refinement of Prototype	11 days	Wed 18-02-26	Wed 04-03-26
Participant Interaction with Prototype	3 days	Thu 05-03-26	Mon 09-03-26
Post-Test Questionnaire	6 days	Tue 10-03-26	Tue 17-03-26
Data Analysis	11 days	Wed 18-03-26	Wed 01-04-26
Complete final thesis	28 days	Thu 02-04-26	Sun 10-05-26



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Appendix

Financial Literacy & Gamification Questionnaire (Pre-Test)

Purpose: This questionnaire is part of a study investigating the impact of a gamified tool on financial literacy and behavior among university students. Your honest responses are crucial to this research. All individual answers will be kept confidential.

Instructions: The questionnaire is divided into eight sections. Please answer all questions.

Section I: Demographics

Please select the option that best describes you.

1. What is your age group?

A. Under 18

B. 18- 24

C. 25–29

D. 30-34

E. 35-39

F. 40 or older

2. Which Faculty or College are you currently enrolled in?

A. Faculty of Engineering and Computing (FENC)

B. Faculty of Science and Sport (FOSS)

C. Faculty of Law (FOL)

D. College of Business and Management (COBAM)

E. College of Health Sciences (COHS)

F. Faculty of The Built Environment

G. Faculty of Education and Liberal Studies

H. Joint College of Medicine, Oral Health & Veterinary Sciences

3. What is your current Level of Study?

A. Diploma

B. Associate Degree

C. Bachelor's Degree

D. Master's Degree

E. Postgraduate Diploma

F. PhD Degree

G. Other

4. What is your current employment status?

A. Full-Time Employed

B. Part-Time Employed

C. Self-Employed / Entrepreneur

D. Unemployed (Actively seeking employment)

E. Not Employed (Full-time student focus)

Section II: General Financial Knowledge

5. How do you rate your general financial knowledge skills?

A. Excellent knowledge and skills

B. Good knowledge and skills

C. Satisfactory knowledge and skills

D. Unsatisfactory knowledge and skills

E. No knowledge and skills

F. Don't Know

6. Of the following steps, which would be the first step in financial planning?

A. Get a job so you can start earning money.

B. Determine your needs and goals for the future.

C. Starting looking for a home to buy.

D. Save a portion of your money for the future.

7. To what extent do you consider yourself knowledgeable in the following areas:	**Excellent knowledge**	**Good knowledge**	**Satisfactory**	**Unsatisfactory**	**None**
How basic investments work eg: stocks, bonds					
How to set up an emergency fund and determine how much to save					
How to differentiate between financial needs and wants					
How to create a monthly budget and track income and expenses					
How diversification reduces investment risk					
How to prioritize and manage repayment of debts					
How inflation affects the value of money saved					

Section III: Saving

8. What is the primary purpose of an Emergency Fund?

A. Buying new electronics on sale.

B. Covering unexpected expenses, like a medical bill or car repair.

C. Investing in the stock market for high returns.

D. Paying for a yearly vacation.

9. Which of the following should you remember when developing a savings plan?

A. Wait until you are 40 years old before saving.

B. Pay yourself first.

C. Pay off your low-interest debt first.

D. Have only high-risk investments.

10. You are following the "Pay Yourself First" savings principle. Which action best reflects this principle?

A. Paying all your bills and then saving whatever money is left over.

B. Transferring a set amount to savings automatically on payday before any spending.

C. Only saving money when you receive an unexpected bonus or gift.

D. Saving a minimum of 10% of your income at the end of the month.

11. Imagine, that you deposited the money in a bank account at 8% interest rate, while the annual inflation rate was 10%. Do you think the money from your account can buy more or less, or the same amount of goods and services on average now as a year ago?

A. More than a year ago

B. The same

C. Less than a year ago

D. Don’t Know

Section IV: Debt Management

12. Which strategy is MOST effective for paying off multiple debts?

A. Paying the smallest debt first to build momentum (Debt Snowball)

B. Ignoring the highest interest rate debts

C. Taking out more loans to cover older loans

D. Don't know

13.Select true or false for the following statements:

	True	False	Don't Know
A debt occurs when you borrow money from someone or a financial institution			
A loan is money borrowed usually from a financial institution and must be paid back with interest			
A credit card is not the same as borrowing money from the bank			

14. A 15-year mortgage typically requires higher monthly payments than a 30-year mortgage but the total interest over the life of the loan will be less.

A. True

B. False

C. Don't Know

15. Suppose you owe $1,000 on a loan and the interest rate you are charged is 20% per year compounded annually. If you didn't pay anything off, at this interest rate, how many years would it take for the amount you owe to double?

A. Less than 2 years

B. 2 to 4 years

C. 5 to 9 years

D. 10 or more years

E. Don't Know

Section V: Investing

16. What is the primary purpose of diversifying your investments?

A. To increase transaction fees.

B. To buy fewer shares of stock.

C. To reduce overall risk.

D. To guarantee a high return.

17. What is the main benefit of starting to invest when you are young?

A. Lower tax rates.

B. More time for compounding to increase returns.

C. Fewer investment options are available.

D. Government subsidies for young investors.

18. If an investment earns "compounding interest," what does that mean?

A. You only earn interest on the original amount you invested.

B. You earn interest on both the original amount and previously earned interest.

C. You have to pay tax on the interest every month.

D. The interest rate decreases over time.

19. You purchase a stock for $50 per share. The company later announces that it is paying a $1 dividend per share to its shareholders. What is this payment called?

A. Capital Gain

B. Dividend

C. Capital Loss

D. Interest Payment

Section VI: Budgeting

20. What is budgeting?

A. The amount of bank transactions made

B. A plan for how to spend money in the future

C. A list of spending in the past

D. Don't know.

21. Alan has created a new budget. Which of the following should he NOT do?

A. Have a spending plan.

B. Spend less than he earns.

C. Use credit for all items not in his budget.

D. Stick to his budget.

22. According to the 50/30/20 budgeting guideline, which category should receive the majority of your monthly income (50%)?

A. Wants (non-essential costs)

B. Needs (important costs like rent, groceries).

C. Savings & debt repayment.

D. Travel and entertainment.

23. Which of the following is considered a need in budgeting?

A. Dining at restaurants

B. Subscription Services

C. Rent or Housing Payments

D. Buying new clothes for enjoyment.

Section VII: Financial Attitude and Behaviour

24. Please indicate how much you agree with the following statements (1=Strongly Disagree, 5=Strongly Agree).

Statement	**1. Strongly Disagree**	**2. Disagree**	**3. Neutral**	**4. Agree**	**5. Strongly Agree**	**N/A**
I feel **confident** in my ability to handle my financial situation.						
I feel **confident** in my ability to make informed financial decisions						
I **track my monthly expenses** and try to stick to a budget.						
I **regularly put money into a savings** or investment account.						
I make **more than the minimum payment** on my debt whenever possible.						

I have **clear financial goals** (e.g., saving for a major purchase, starting to invest).						
I **compare financial products** or services before making a financial decision						
I **seek advice** before making an important financial decision						

Section VIII: Gamification Effectiveness

25. Please indicate how much you agree with the following statements regarding the gamified tool you used (1=Strongly Disagree, 5=Strongly Agree).

Statement	**1. Strongly Disagree**	**2. Disagree**	**3. Neutral**	**4. Agree**	**5. Strongly Agree**
A game-based tool will be **more engaging** than traditional financial lessons.					
A gamified format will make complex financial concepts **easier to understand**.					
The low-stakes environment of a game can help **reduce my anxiety** about learning finance.					
A **points, badges, or rewards** system can motivate me to complete learning modules.					
A gamified learning tool can motivate me to **apply learned**					

knowledge to my real-life savings or budgeting habits.					

Financial Literacy & Gamification Questionnaire (Post-Test)

Purpose: This questionnaire is part of a study investigating the impact of a gamified tool on financial literacy and behavior among university students. Your honest responses are crucial to this research. All individual answers will be kept confidential.

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F. Faculty of The Built Environment

G. Faculty of Education and Liberal Studies

H. Joint College of Medicine, Oral Health & Veterinary Sciences

3. What is your current Level of Study?

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How to set up an emergency fund and determine how much to save					
How to differentiate between financial needs and wants					
How to create a monthly budget and track income and expenses					
How diversification reduces investment risk					
How to prioritize and manage repayment of debts					
How inflation affects the value of money saved					

Section III: Saving

8. What is the primary purpose of an Emergency Fund?

A. Buying new electronics on sale.

B. Covering unexpected expenses, like a medical bill or car repair.

C. Investing in the stock market for high returns.

D. Paying for a yearly vacation.

9. Which of the following should you remember when developing a savings plan?

A. Wait until you are 40 years old before saving.

B. Pay yourself first.

C. Pay off your low-interest debt first.

D. Have only high-risk investments.

10. You are following the "Pay Yourself First" savings principle. Which action best reflects this principle?

A. Paying all your bills and then saving whatever money is left over.

B. Transferring a set amount to savings automatically on payday before any spending.

C. Only saving money when you receive an unexpected bonus or gift.

D. Saving a minimum of 10% of your income at the end of the month.

11. Imagine, that you deposited the money in a bank account at 8% interest rate, while the annual inflation rate was 10%. Do you think the money from your account can buy more or less, or the same amount of goods and services on average now as a year ago?

A. More than a year ago

B. The same

C. Less than a year ago

D. Don't Know

Section IV: Debt Management

12. Which strategy is MOST effective for paying off multiple debts?

A. Paying the smallest debt first to build momentum (Debt Snowball)

B. Ignoring the highest interest rate debts

C. Taking out more loans to cover older loans

D. Don't know

13.Select true or false for the following statements:

	True	False	Don’t Know
A debt occurs when you borrow money from someone or a financial institution			
A loan is money borrowed usually from a financial institution and must be paid back with interest			
A credit card is not the same as borrowing money from the bank			

14. A 15-year mortgage typically requires higher monthly payments than a 30-year mortgage but the total interest over the life of the loan will be less.

A. True

B. False

C. Don't Know

15. Suppose you owe $1,000 on a loan and the interest rate you are charged is 20% per year compounded annually. If you didn't pay anything off, at this interest rate, how many years would it take for the amount you owe to double?

A. Less than 2 years

B. 2 to 4 years

C. 5 to 9 years

D. 10 or more years

E. Don't Know

Section V: Investing

16. What is the primary purpose of diversifying your investments?

A. To increase transaction fees.

B. To buy fewer shares of stock.

C. To reduce overall risk.

D. To guarantee a high return.

17. What is the main benefit of starting to invest when you are young?

A. Lower tax rates.

B. More time for compounding to increase returns.

C. Fewer investment options are available.

D. Government subsidies for young investors.

18. If an investment earns "compounding interest," what does that mean?

A. You only earn interest on the original amount you invested.

B. You earn interest on both the original amount and previously earned interest.

C. You have to pay tax on the interest every month.

D. The interest rate decreases over time.

19. You purchase a stock for $50 per share. The company later announces that it is paying a $1 dividend per share to its shareholders. What is this payment called?

A. Capital Gain

B. Dividend

C. Capital Loss

D. Interest Payment

Section VI: Budgeting

20. What is budgeting?

A. The amount of bank transactions made

B. A plan for how to spend money in the future

C. A list of spending in the past

D. Don't know.

21. Alan has created a new budget. Which of the following should he NOT do?

A. Have a spending plan.

B. Spend less than he earns.

C. Use credit for all items not in his budget.

D. Stick to his budget.

22. According to the 50/30/20 budgeting guideline, which category should receive the majority of your monthly income (50%)?

A. Wants (non-essential costs)

B. Needs (important costs like rent, groceries).

C. Savings & debt repayment.

D. Travel and entertainment.

23. Which of the following is considered a need in budgeting?

A. Dining at restaurants

B. Subscription Services

C. Rent or Housing Payments

D. Buying new clothes for enjoyment.

Section VII: Financial Attitude and Behaviour

24. Please indicate how much you agree with the following statements (1=Strongly Disagree, 5=Strongly Agree).

Statement	**1. Strongly Disagree**	**2. Disagree**	**3. Neutral**	**4. Agree**	**5. Strongly Agree**	**N/A**
I feel **confident** in my ability to handle my financial situation.						
I feel **confident** in my ability to make informed financial decisions						
I **track my monthly expenses** and try to stick to a budget.						
I **regularly put money into a savings** or investment account.						
I make **more than the minimum payment** on my debt whenever possible.						

I have **clear financial goals** (e.g., saving for a major purchase, starting to invest).						
I **compare financial products** or services before making a financial decision						
I **seek advice** before making an important financial decision						

Section VIII: Gamification Effectiveness

25. Please indicate how much you agree with the following statements regarding the gamified tool you used (1=Strongly Disagree, 5=Strongly Agree).

Statement	**1. Strongly Disagree**	**2. Disagree**	**3. Neutral**	**4. Agree**	**5. Strongly Agree**
The game-based tool was **more engaging** than traditional financial lessons.					
The gamified format made complex financial concepts **easier to understand.**					
The low-stakes environment of the game helped **reduce my anxiety** about learning finance.					
The **points, badges, or rewards** motivated me to complete the learning modules.					
After using the game, I intend to **apply the knowledge to my**					

real-life savings or budgeting habits.					

Interview questions for experts

1. What is your definition of a student who is financially literate?
2. In your estimation, how do you assess the current level of financial literacy among university students?
3. What are the financial concepts that students struggle with the most?
4. How important do you think financial literacy is for all university students?
5. What are the most important financial concepts that should be taught across the board?
6. Have you observed the financial literacy level increase or decrease throughout your teaching career?
7. What gaps or limitations do you see in traditional teaching methods on finance?
8. Do you think the current teaching methodology is sufficiently engaging students in the area of study?
9. Do you think a gamified approach to financial literacy could be beneficial to students?
10. Have you ever used elements of gamification in your teaching?
11. What elements of gamification would be best to include in a gamified financial literacy platform?
12. Which of the following aspects of managing personal finances do you believe should necessarily be part of a financial training course for people:

Aspects	YES	NO	Don't Know
How to form one's own financial targets and draw a personal current financial plan			
Mortgage Loans			
Insurance and insurance products			
What information should a user pay attention to when signing a contract with a bank or another financial company			
Private pension funds			
Consumer credits to purchase goods and services			
What to do not to become overwhelmed with debt			
Capital markets, stock and unit fund shares			
How to plan purchases of durables (car, apartment, dacha) and evaluate one's abilities to implement them			